

核心思路重新梳理

要解决的不只是 single-shot evaluation 的问题，而是要建立一个 comprehensive benchmark that addresses **two critical dimensions**:

- 1. **Realistic Data Composition:** 通过调整 spam ratio 来模拟 real-world scenarios
- 2. **Synthetic Data Integration:** 通过调整 synthetic ratio 来找到 optimal mixing strategies

实验设计的两个关键维度

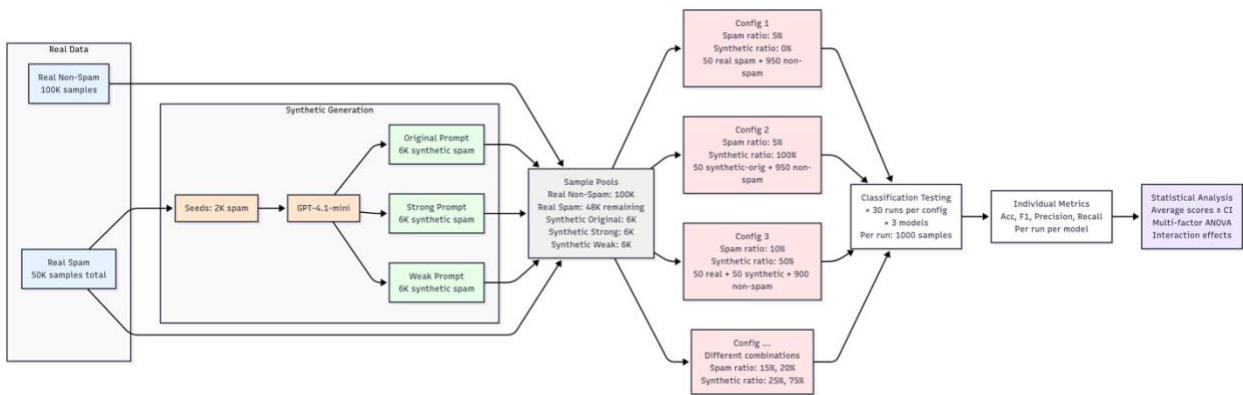
基于我的理解，实验设计应该包含：

Dimension 1: Spam Ratio Control

- 5% spam: 50 malicious + 950 benign = 1000 total
- 10% spam: 100 malicious + 900 benign = 1000 total
- 15% spam: 150 malicious + 850 benign = 1000 total
- 20% spam: 200 malicious + 800 benign = 1000 total

Dimension 2: Synthetic Mixing Ratio (within malicious samples)

- 0% synthetic: 100% real malicious
- 25% synthetic: 25 synthetic + 75 real malicious
- 50% synthetic: 50 synthetic + 50 real malicious
- 75% synthetic: 75 synthetic + 25 real malicious
- 100% synthetic: 100% synthetic malicious



实验规模估算

基于这种设计，总实验规模为：

- 4 spam ratios × 5 synthetic ratios × 3 prompt strategies × 3 models × 30 runs
- **总计:** 5,400 individual experiments

典型 Configuration 示例

1. **Pure Real Baseline:** 10% spam (100 real malicious + 900 real benign)
2. **Pure Synthetic:** 10% spam (100 synthetic-original + 900 real benign)
3. **Balanced Mix:** 10% spam (50 real + 50 synthetic-original + 900 real benign)
4. **Multi-Strategy:** 10% spam (33 original + 33 strong + 34 weak + 900 real benign)

我建议的具体实施设计

Data Pipeline Architecture

基于现有的 batch6 架构，我建议以下实施方案：

数据准备:

- 利用现有的 2K seeds 和已生成的 18K synthetic samples
- 从 CEAS-08 剩余的 48K real spam 和 100K benign 构建 sampling pools
- 为 multi-sample evaluation 设计 stratified sampling protocol

实验配置管理:

Configuration Matrix:

- 每个 config 由(spam_ratio, synthetic_ratio, strategy, model)唯一标识
- 每个 config 重复 30 次 runs
- 每次 run 生成 1000 个 test samples
- 使用 pre-trained models 避免重复训练成本

统计分析框架:

- 每个 configuration: 30 个 F1 scores → mean ± CI
- Cross-configuration comparisons: paired t-tests
- Multi-factor ANOVA for interaction effects
- Effect size analysis for practical significance

分阶段实施策略

Phase 1: Pilot Study

- 限制 scope: 2 spam ratios × 3 synthetic ratios × 2 strategies × 1 model × 5 runs
- 总计: 60 experiments
- 目的: validate methodology 和 identify key factors

Phase 2: Selective Expansion

- 基于 pilot results, expand 到 most promising combinations
- 目的: focus on configurations with largest effect sizes

Phase 3: Full Deployment

- 仅在 proven value 后进行 complete experimental matrix

🤔 我的关键疑问和担心

疑问 1: 实验规模的可行性

5,400 个 experiments 的规模让我担心:

- 计算时间和成本是否现实?
- 数据管理和结果 tracking 的复杂度?
- 是否需要 cloud computing infrastructure?

疑问 2: Statistical Design 的优化

- 是否可以用 fractional factorial design 减少 experiments?
- 如何处理 multiple comparisons 问题?
- Response surface methodology 是否更适合这种 multi-dimensional optimization?

疑问 3: 实际价值的评估

- 在 CEAS-08 上找到的 optimal ratios 能否推广到其他 cybersecurity datasets?
- 这种 highly specific 的 tuning 是否有 practical deployment value?
- 投入这么大规模的实验, expected insights 的价值如何量化?

疑问 4: 技术实施细节

- Multi-run 的 sampling strategy: 是否需要 ensure no overlap between runs?
- Different configurations 在同一 run 中是否应该使用 same benign samples?
- Pre-trained models 的 consistency across different data compositions?

疑问 5: 结果解释的复杂性

- Multi-way ANOVA results 如何 interpret for practical guidance?
- 如何处理 potential interaction effects between factors?
- 最终如何提供 actionable recommendations for practitioners?